Ayush Kumar

AI Engineer | LLM & Agentic Systems

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Profile Summary

AI engineer shipping LLM systems at scale: a warranty adjudication pipeline saving **$1M/month** and an in-house CRM used by **4,000+ technicians**. Strong in agentic RAG, LLM evaluation, multi-agent orchestration, and Python backends.

Education

Indian Institute of Technology Guwahati, B.Tech CSE; Minor: Data Science and AI — Aug 2022 – Jun 2026 | CGPA: 8.92/10

Experience

Ola Electric — Jun 2026 – Present
Graduate Engineer Trainee, Applied AI — Bengaluru

AI Warranty Adjudication Pipeline

- Closed a fraud loop where technicians claimed replacements for healthy parts and resold the originals: evidence-based auto-rejection cut daily claims from **1,500 to 500**, saving **$1M/month**.
- Built an LLM system that approves/rejects claims against part-specific SOPs, reaching **90%** accuracy vs human reviewers.
- Built a self-improving prompt pipeline that lifted accuracy from **86% to 90%**: human audit corrections on model errors auto-generate targeted rule and prompt patches, validated via golden-set regression and shadow traffic.

SAFAI (In-House CRM, Salesforce Replacement)

- Co-built a Spring Boot Salesforce replacement in 10 days, preserving access to **5M+** historical job cards and covering tickets, inventory, billing, insurance, and pan-India role-based access control.
- Primary engineer for warranty and insurance integration with Reliance, Acko, and Ola.
- Built an MCP war-room agent that diagnoses issues from code and DB and raises fix PRs, fixing **3,400** job cards.
- Supported national rollout, onboarding **4,000+** technicians and processing 5,000 job cards and 25K parts daily.

Ola Krutrim — May 2025 – Jul 2025
AI Engineer Intern, Ola Maps — Bengaluru

- Trained a spaCy NER model for Ola Maps autocomplete (**50M+ daily API calls**), lifting top-3 search accuracy by **2%**.
- Built ETA prediction models tuned with Bayesian Optimization on geospatial and temporal features, and clustered LBS wayIds into sub-city ride zones with DBSCAN, cutting MAE by **3 min** in low-data tier-3 regions.

Undergraduate Research, IIT Guwahati — Aug 2025 – May 2026
Agentic Argument Mining; Advisor: Prof. Ashish Anand — Report

- Designed a 3-agent pipeline (extraction, relation mapping, critic) with iterative self-correction for argument mining.
- Built a hybrid retriever: ChromaDB semantic search + DMRST discourse parsing + spectral graph similarity (NetworkX).
- With GPT-5 Mini and the agentic pipeline alone, **outperformed fine-tuned SOTA baselines (TANL, DENIM) on all four argument component and relation metrics** (e.g., ACI 0.649 vs 0.621 for DENIM).
- Scaled the ARC policy corpus from 81 to 444 paragraphs via LLM-assisted annotation (relation F1: 4.76 to 25.00).
- Co-authored the resource paper on corpus extension and agentic annotation, submitted to **LREC 2026**.

Projects

LLM-Compiler-Based Agentic RAG System for Legal QA — Nov 2024 – Dec 2024
13th Inter-IIT Tech Meet, IIT Bombay — GitHub

- Built an agentic RAG system on LLM-Compiler with DAG-based orchestration, dynamic replanning, and parallel tool calls.
- Optimized legal retrieval with Voyage-Law2, auto-merging recursive chunking, Cohere reranking, and a fine-tuned beam retriever, with metadata-aware citations and Claude as fallback LLM. Reached **86%** on CUAD and **84%** on AILA.

Multi-Agent Medical Report Analysis — Jan 2026 – Feb 2026
Kaggle MedGemma Impact Challenge — GitHub

- Built a 14-agent MedGemma system (Vertex AI) that extracts patient info, medications, lab values, and imaging findings from medical PDFs, with interactive Q&A over reports. Shipped with FastAPI, React, and Docker.
- Annotated a custom 301-image dataset and fine-tuned YOLOv8 to detect X-ray, CT, MRI, and ultrasound images in PDFs.

Achievements

- **Inter-IIT Tech Meet: Gold medal / Rank 1** of 20 IITs (Mphasis, 12th); **Rank 4** of 22 IITs (Pathway GenAI, 13th).
- **Competitions: AIR 2**, GeeksforGeeks EROH scholarship (10k+ students of IITs, NITs, IIITs); **3rd**, OES ML Challenge.
- **Codeforces:** Max rating **1727** (mahnameisayush). **Olympiads:** Physics state topper; KVPY Fellow (AIR 863).

Technical Skills

Languages: Python, C/C++, SQL
LLM & Agents: LangGraph, LangChain, MCP, RAG, tool calling, LLM evaluation, prompt optimization, ChromaDB
ML & Backend: PyTorch, spaCy, scikit-learn, pandas, NumPy, YOLOv8, FastAPI, REST APIs, Redis, Docker